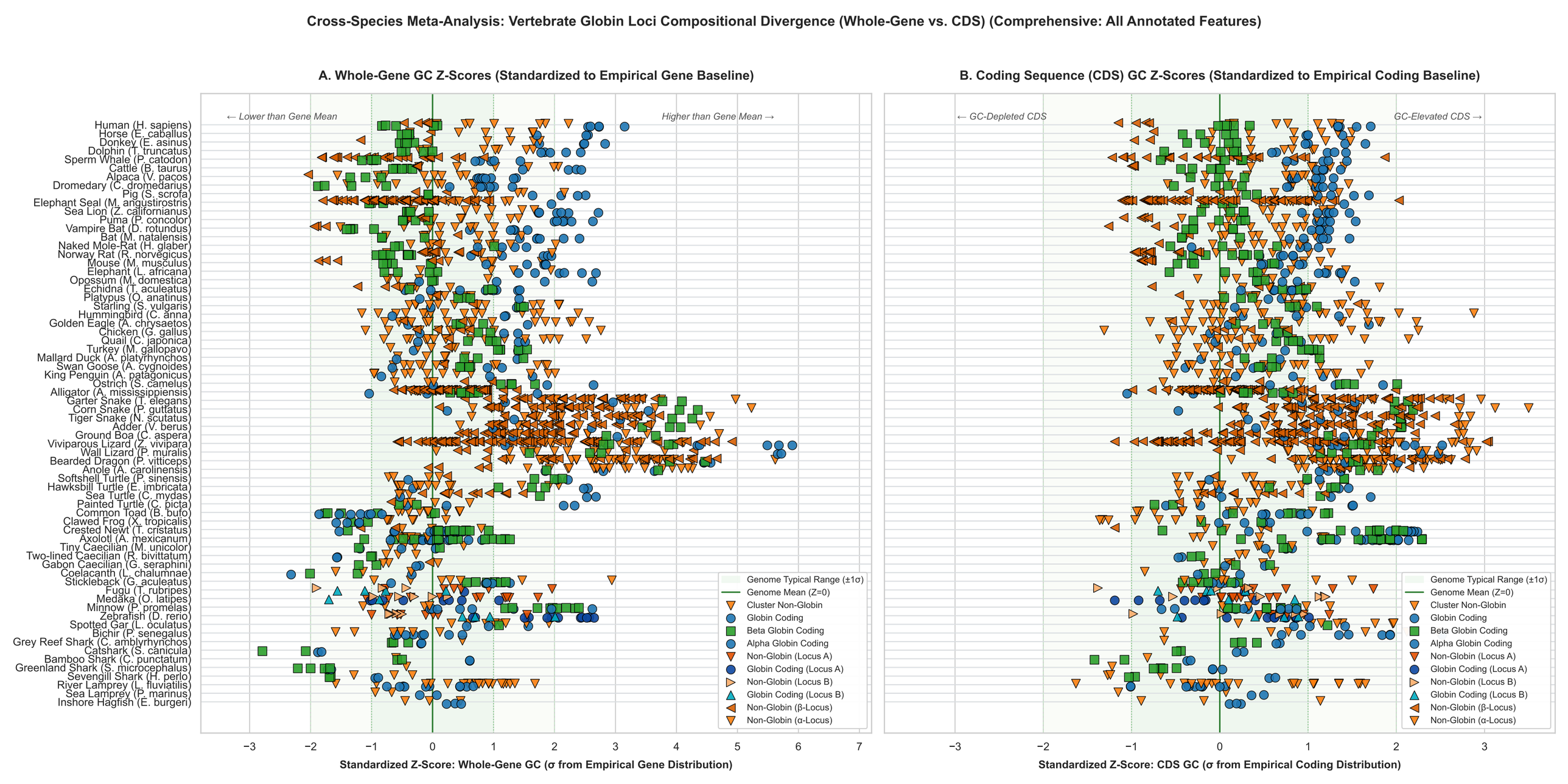

Cross-Species Meta-Analysis: Vertebrate Globin Loci Compositional Divergence (Whole-Gene vs. CDS) (Comprehensive: All Annotated Features)
A. Whole-Gene GC Z-Scores (Standardized to Empirical Gene Baseline)
B. Coding Sequence (CDS) GC Z-Scores (Standardized to Empirical Coding Baseline)
← Lower than Gene Mean
Higher than Gene Mean →
← GC-Depleted CDS
GC-Elevated CDS →
Human (H. sapiens)
Horse (E. caballus)
Donkey (E. asinus)
Dolphin (T. truncatus)
Sperm Whale (P. catodon)
Cattle (B. taurus)
Alpaca (V. pacos)
Dromedary (C. dromedarius)
Pig (S. scrofa)
Elephant Seal (M. angustirostris)
Sea Lion (Z. californianus)
Puma (P. concolor)
Vampire Bat (D. rotundus)
Bat (M. natalensis)
Naked Mole-Rat (H. glaber)
Norway Rat (R. norvegicus)
Mouse (M. musculus)
Elephant (L. africana)
Opossum (M. domestica)
Echidna (T. aculeatus)
Platypus (O. anatinus)
Starling (S. vulgaris)
Hummingbird (C. anna)
Golden Eagle (A. chrysaetos)
Chicken (G. gallus)
Quail (C. japonica)
Turkey (M. gallopavo)
Mallard Duck (A. platyrhynchos)
Swan Goose (A. cygnoides)
King Penguin (A. patagonicus)
Ostrich (S. camelus)
Alligator (A. mississippiensis)
Garter Snake (T. elegans)
Corn Snake (P. guttatus)
Tiger Snake (N. scutatus)
Adder (V. berus)
Ground Boa (C. aspera)
Viviparous Lizard (Z. vivipara)
Wall Lizard (P. muralis)
Bearded Dragon (P. vitticeps)
Anole (A. carolinensis)
Softshell Turtle (P. sinensis)
Hawksbill Turtle (E. imbricata)
Sea Turtle (C. mydas)
Painted Turtle (C. picta)
Common Toad (B. bufo)
Clawed Frog (X. tropicalis)
Crested Newt (T. cristatus)
Axolotl (A. mexicanum)
Tiny Caecilian (M. unicolor)
Two-lined Caecilian (R. bivittatum)
Gabon Caecilian (G. seraphini)
Coelacanth (L. chalumnae)
Stickleback (G. aculeatus)
Fugu (T. rubripes)
Medaka (O. latipes)
Minnow (P. promelas)
Zebrafish (D. rerio)
Spotted Gar (L. oculatus)
Bichir (P. senegalus)
Grey Reef Shark (C. amblyrhynchos)
Catshark (S. canicula)
Bamboo Shark (C. punctatum)
Greenland Shark (S. microcephalus)
Sevengill Shark (H. perlo)
River Lamprey (L. fluviatilis)
Sea Lamprey (P. marinus)
Inshore Hagfish (E. burgeri)
−3
−2
−1
0
1
2
3
4
5
6
7
Standardized Z-Score: Whole-Gene GC (σ from Empirical Gene Distribution)
Standardized Z-Score: CDS GC (σ from Empirical Coding Distribution)
Genome Typical Range (±1σ)
Genome Mean (Z=0)
Cluster Non-Globin
Globin Coding
Beta Globin Coding
Alpha Globin Coding
Non-Globin (Locus A)
Globin Coding (Locus A)
Non-Globin (Locus B)
Globin Coding (Locus B)
Non-Globin (β-Locus)
Non-Globin (α-Locus)